

SHUBHAM PATTHE

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PROFESSIONAL SUMMARY

ML Engineer with research experience in NLP and deep learning at IIT Kharagpur. Built production-ready systems including a FinBERT-based fraud detector (96% recall) and RAG pipelines. Proficient in PyTorch, LangChain, and end-to-end model deployment.

EDUCATIONS

Indian Institute Of Technology, Kharagpur | Kharagpur, India

Nov 2022 - Present

Dual Degree(B.tech. + M.Tech.)

WORK EXPERIENCES

Data Science Intern | Internship

Dec 2025 - Jan 2026

WeIntern

Remote • Pune

- **Built** and cleaned large datasets of **10,000+ records** using **Python** (Pandas, NumPy), **reducing data preprocessing time by ~30%** and enhancing model input quality.
- **Conducted** exploratory data analysis (EDA) on structured financial and text datasets, **uncovering 5+ key trends and correlations** that informed feature selection and model design.
- **Engineered** a performance analytics system processing **600+ weekly intern records** across 5 domains, **implementing a composite scoring algorithm** and generating **6+ statistical visualizations** using **Apache Superset**, Pandas, and Matplotlib.

Technologies / Skills Used : Pandas , Numpy, Apache Superset

PROJECTS

Fraud Detection | **Prof. C.S. Mishra** |  Github

Jan 2026 - Apr 2026

BTP Project

- Developed a **labelled dataset** comprising 7,595 earnings-call transcript segments from 12 companies (6 SEC-fraud, 6 controls) spanning 2003-2021, implementing document-level splits to mitigate data leakage.
- Fine-tuned **ProsusAI/FinBERT** for **binary fraud classification**, achieving 84% accuracy and 89% fraud recall; enhanced it into a **Hybrid Detector** by integrating FinBERT embeddings with 6 linguistic deception features, raising fraud recall to 96% and macro-F1 score to 87%.
- Conducted an early-detection test on a pre-fraud GE transcript (Q1 2015), resulting in a 97.78% fraud probability—more than a year prior to SEC enforcement—validating its real-world screening utility.

Technologies / Tools Used : Python, NLP

Measuring Corporate Culture

Jul 2025 - Nov 2025

B.Tech Project

- **Computed** culture scores from 60+ earnings call transcripts from HCL Technologies using **semi-supervised NLP pipelines**.
- **Achieved** RMSE ≈ 0.023 and MAE ≈ 0.018 against benchmark scores.
- **Cleaned** and processed transcripts, constructed culture dictionaries, and trained **Word2Vec embeddings** to quantify dimensions such as teamwork, innovation, integrity, respect, and quality.
- **Visualized** cultural clusters with t-SNE.

Technologies / Tools Used : NLP, matplotlib, Python, pandas, numpy, scikit-learn, gensim (Word2Vec), seaborn

LLM-Powered Document Intelligence System

Nov 2025 - Feb 2026

Self Project

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline enabling natural language Q&A over custom document corpora, achieving <2s average response latency on Hugging Face Spaces.
- Implemented semantic chunking and embedding-based retrieval using sentence-transformers and ChromaDB, improving context relevance by ~40% over naive keyword search baselines.
- Evaluated pipeline quality using the RAGAS framework across faithfulness, answer relevancy, and context precision metrics on 200+ test queries, iteratively tuning chunk size and retrieval parameters to raise faithfulness score from 0.71 to 0.89.
- Deployed as an interactive Streamlit application with source citation and confidence scoring, enabling traceable, hallucination-reduced outputs.

Technologies / Tools Used : Python, LangChain, ChromaDB, RAGAS, Groq (LLaMA-3), Streamlit, Hugging Face Spaces

SKILLS

Machine Learning Deep Learning

: PyTorch, TensorFlow, Scikit-learn, FinBERT, Word2Vec, FAISS, LangChain, HuggingFace

Nlp :

Text Classification, Embeddings, RAG, Fine-tuning, Sentiment Analysis, Named Entity Recognition

Data :

Pandas, NumPy, Apache Superset, Matplotlib, Seaborn

Databases :

SQL, ChromaDB (Vector DB)

Tools :

Git, Streamlit, Hugging Face Spaces, VS Code

CERTIFICATIONS

Algozenieth DSA and CP Course